One-Period Macroeconomic Model

Javier Tasso

Introduction

- General equilibrium model where markets clear and agents optimize subject to their constraints.
  1. Consumers: Maximize utility through the trade-off between consumption and leisure.
  2. Firms: Maximize profits based on their production technology.
  3. Government: Provides public goods through taxation.
- Understand how fiscal policy fundamentally affects output, employment, and real wages.
- No saving, borrowing, or investment. Why?
- All variables are real.

Summary

The Representative Consumer

The Representative Firm

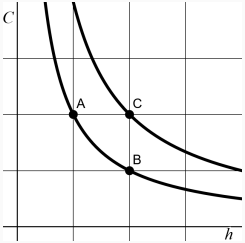
The Government

Equilibrium

Changes in the Model

The Representative Consumer

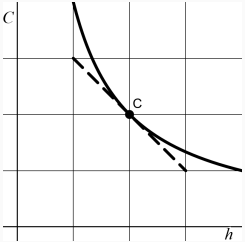
Preferences



$$U(C, h)$$

- Preferences over consumption C and leisure h represented by an utility function.
- Indifference curves.
- Properties.

Marginal Rate of Substitution

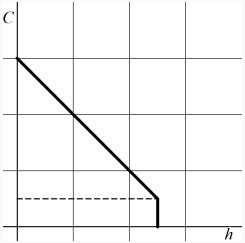


$$\text{MRS}_{h,C}$$

- The (absolute value of the) slope of the indifference curve.
- The rate at which the consumer is just willing to substitute leisure for consumption goods.
- Properties.

Budget Constraint

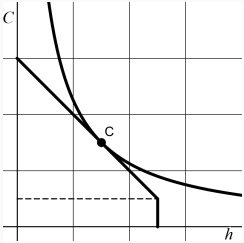
$$C = wL + \Pi - T$$



- Leisure, labor, and total time.
- Change w (real wage).
- Change Π and/or T.
- Normalization $p = 1$.

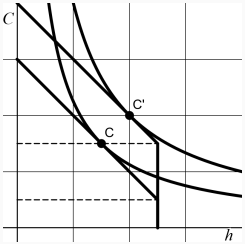
Optimal Choice

$$\text{MRS}_{h,C} = w$$



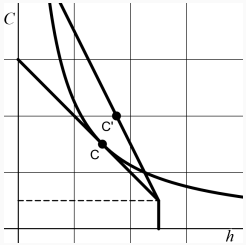
- Reach the highest indifference curve available.
- In the optimal choice the marginal rate of substitution equals the real wage. Why? Assume not.

Effect of Lower Taxes: ↓T



- Parallel shift of the budget constraint.
- Because both consumption and leisure are normal goods, the agent:
  - Consumes more.
  - Works less.
- A change in profits Π produces similar effects.
- Slope of the budget constraint did not change.

Effect of Higher Wage: ↑w



- Slope changes.
- Because of the income and substitution effect, it's not possible to anticipate if the consumer will work more or not.
  - Substitution effect. A higher wage incentivizes the consumer to work more.
  - Income effect. A richer consumer wants more leisure.
- Their sizes will determine the overall effect.

Labor Supply

- Typically we assume the substitution effect dominates.
- Labor supply slopes up.
- Movements along the same labor supply.
- Shifts of labor supply:
  - Crucially, taxes and profits will shift the labor supply.
- How elastic is labor supply? Empirical question.

The Representative Firm

The Production Function

$$Y = A \cdot f(K, L)$$

- Y: Real output per period.
- K: Capital inputs used per period, fixed in the short run.
- L: Labor inputs used per period.
- A: Total factor productivity.

The Production Function in the Short Run

$$Y = A \cdot f(\bar{K}, L)$$

1. K is fixed.
2. Increases with labor: $\mathrm{MP_L}$ is positive.
3. Exhibits diminishing returns: $\mathrm{MP_L}$ is positive, but decreases.
4. Shifts up if capital increases or if total factor productivity increases.

Profit Maximization: Labor Demand

$$\mathrm{MP_L} = w$$

- Optimal choice for the firm. A restatement of the condition $\mathrm{MR} = \mathrm{MC}$.
- Consider cases when $\mathrm{MP_L} \neq w$.
- Total factor productivity or capital shift the demand for labor.
- Graph.

Profits

- Total revenue minus total cost.
- The firm chooses the amount of labor that maximizes its profits.
- The consumer is the owner of the firm.
- Profits are distributed to the consumer.
- That's why they show up in the consumer budget constraint.

The Government

Budget Constrait

$$T = G$$

- The government raises taxes and spends the money G.
- No debt, why?
- In general. No investment or savings, why?

Equilibrium

The Model

- Exogenous variables:
  - Government spending G.
  - Total factor productivity, A.
  - The stock of capital, $\bar{K}$.
- Endogenous variables:
  - Consumption, C.
  - Real wage, w. Amount of labor.
  - Output.
  - Taxes.

Markets and Prices

1. Goods market.
   - Price is normalized $p = 1$.
   - Firms supply output Y, consumers and government demand C and G.
2. Labor market.
   - Price is the real wage w.
   - Firms demand labor and consumer supply it.

Equilibrium Definition

A competitive equilibrium is a set of endogenous quantities (C, L_S, L_D, T, and Y) and an endogenous real wage w, such that given exogenous G, A, $\bar{K}$:

1. The consumer chooses C and L_S maximizing his utility given w, T, and Π.
2. The firm chooses L_D maximizing its profits given w, A, and $\bar{K}$.
3. The labor market clears $L_S = L_D$.
4. The gov't budget constraint is satisfied $T = G$.

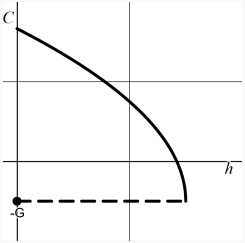
Income-Expenditure Identity

$$Y = C + G$$

- Why? Work with the budget constraints and incorporate the profits.
- Because $Y = A \cdot f(\bar{K}, L) = A \cdot f(\bar{K}, 24 - h)$, there's a relationship between consumption and leisure.
- Visual representation of the model using consumption and leisure (similar to what we did with the representative consumer).

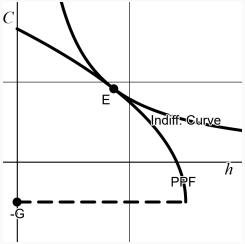
Production Possibilities Frontier

$$C = A \cdot f(\bar{K}, 24 - h) - G$$



- If the consumer does no work, $C = -G$.
- If the consumer works 24 hours a day, maximum level of output.
- Why convex? Labor has decreasing returns.

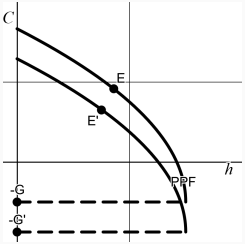
Equilibrium



- The competitive equilibrium coincides to what the consumer would do if she max. her utility subject to the PPF.
- This graph shows all the endogenous variables in equilibrium.
- Where's the real wage in this graph?

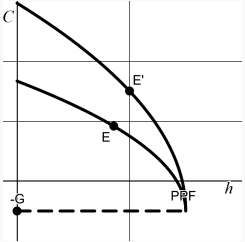
Changes in the Model

Fiscal Policy ↑G



- Consumption and leisure decrease. Why?
- What happens with output?
- Recall $Y = C + G$. Private consumption is partially crowded out.
- Real wage falls. Why? Check slope.
- Expansionary fiscal policy increases output and employment.

Total Factor Productivity ↑A



- Consumption increases. Leisure undetermined. Why?
- What happens with output?
- Real wage increases. Why? Check slope.
- An increase in TFP increases output & consumption, real wages, and likely employment.

Comments

- Microfounded model.
  - Explicit mention to consumer preferences.
  - Explicit mention to technology.
  - Explicit treatment of the budget constraint. Consumer and government.
- Can't say anything about savings, investment, and government debt.
  - Because there's no future in this model.
- Real model. All variables are real.

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